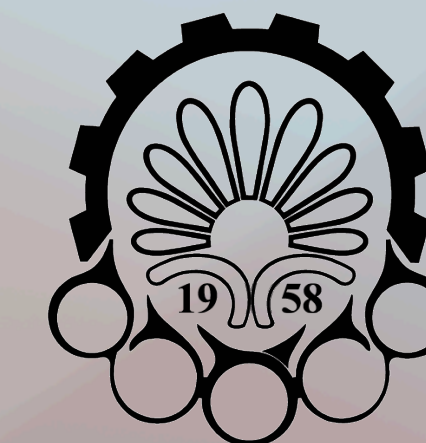


Neural Networks That Remember: Modeling Sequential Data with RNNs

Computation Intelligence and its Application in Mechatronics

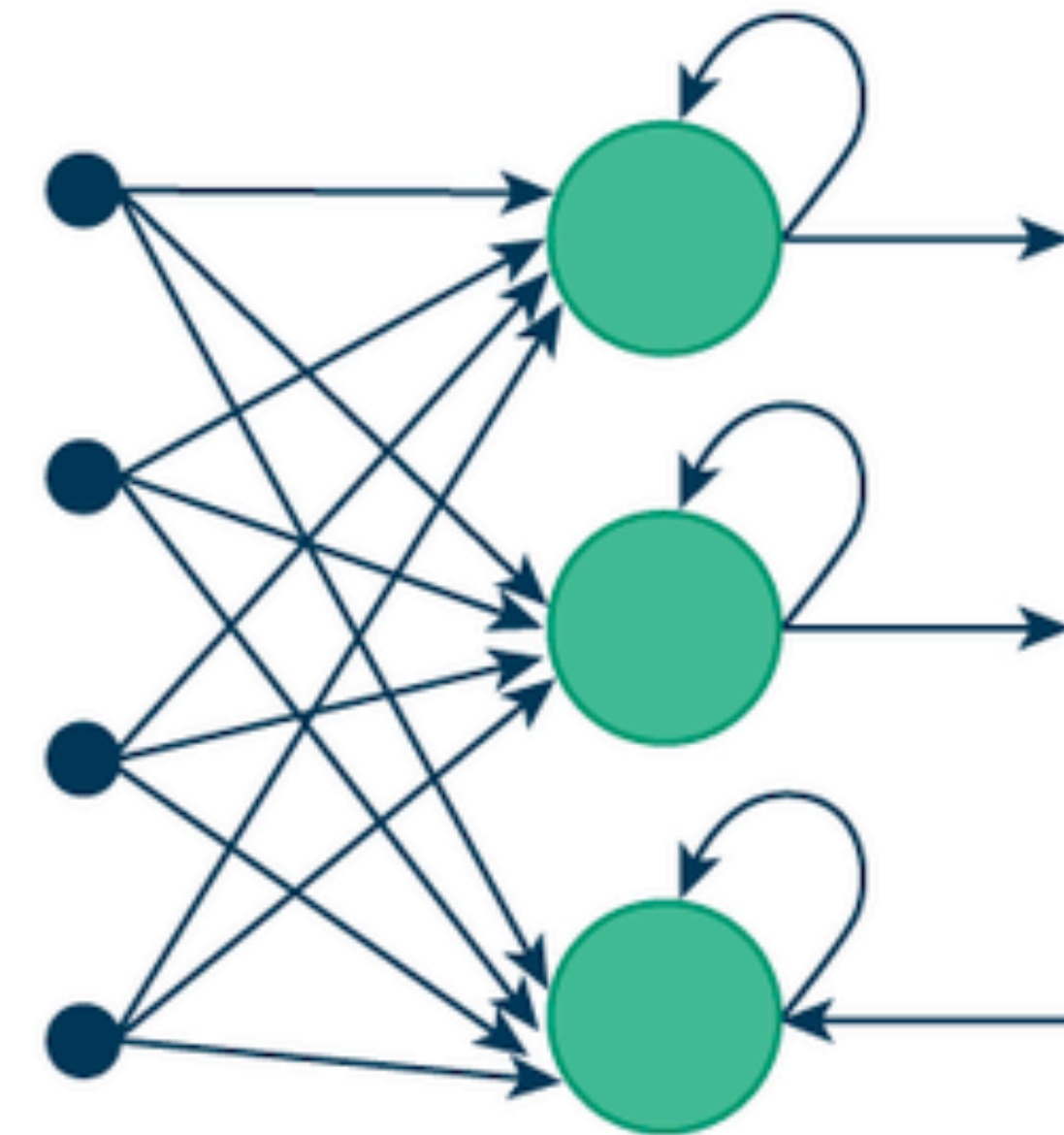
Spring 2025



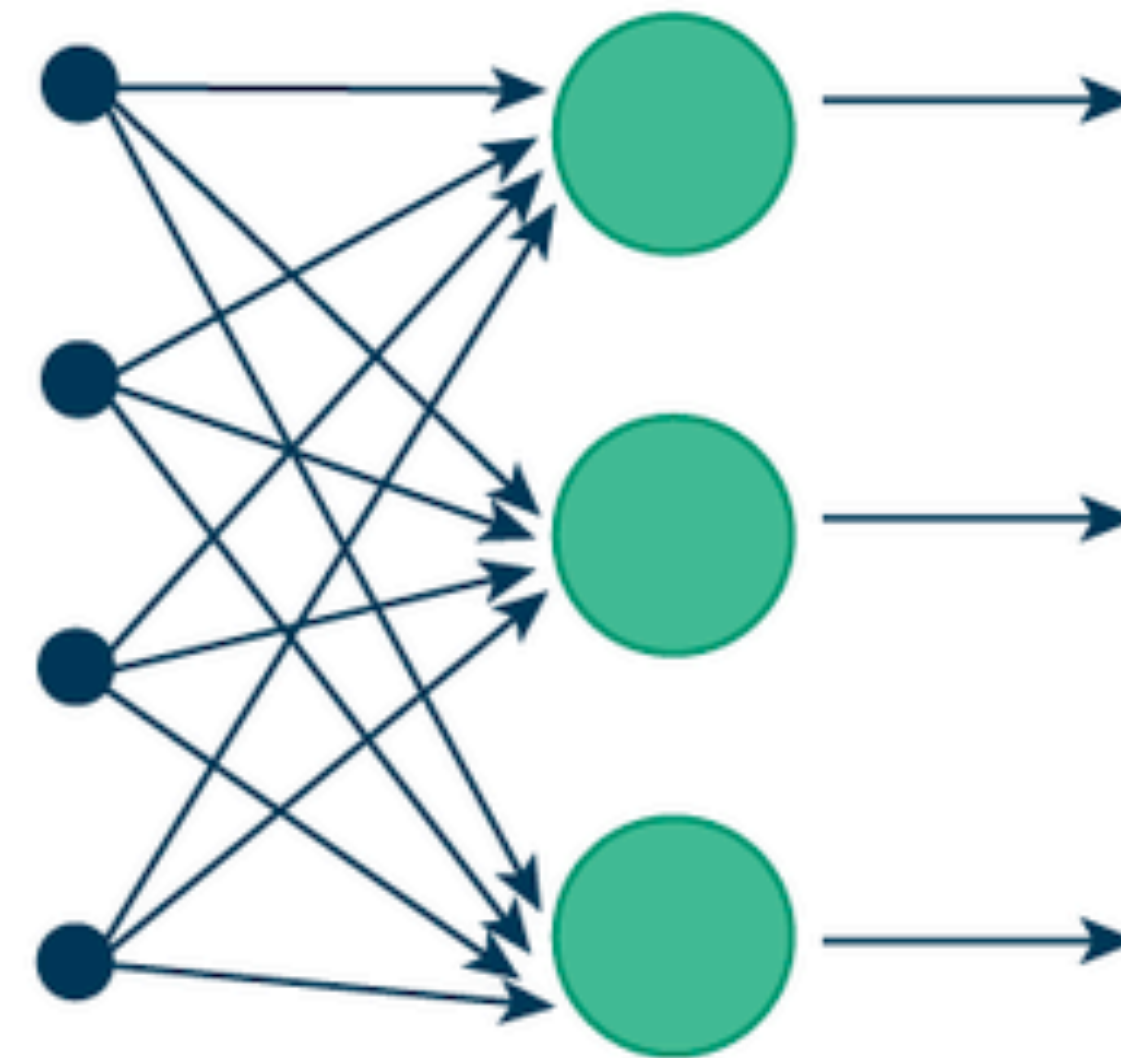
Amirkabir University of Technology
(Tehran Polytechnic)

RNN Fundamentals

MLP vs. RNN



(a) Recurrent Neural Network

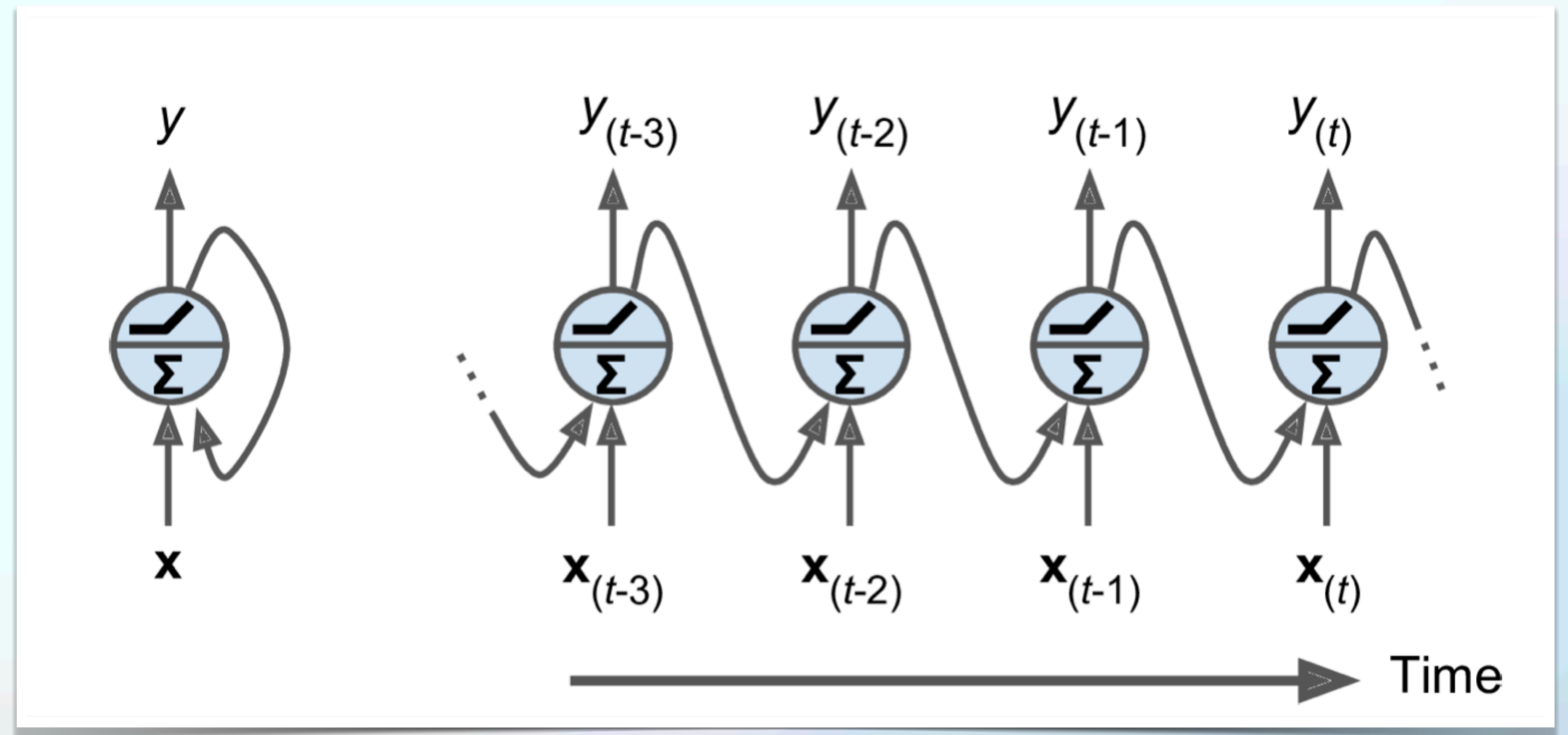


(b) Feed-Forward Neural Network

RNN Fundamentals

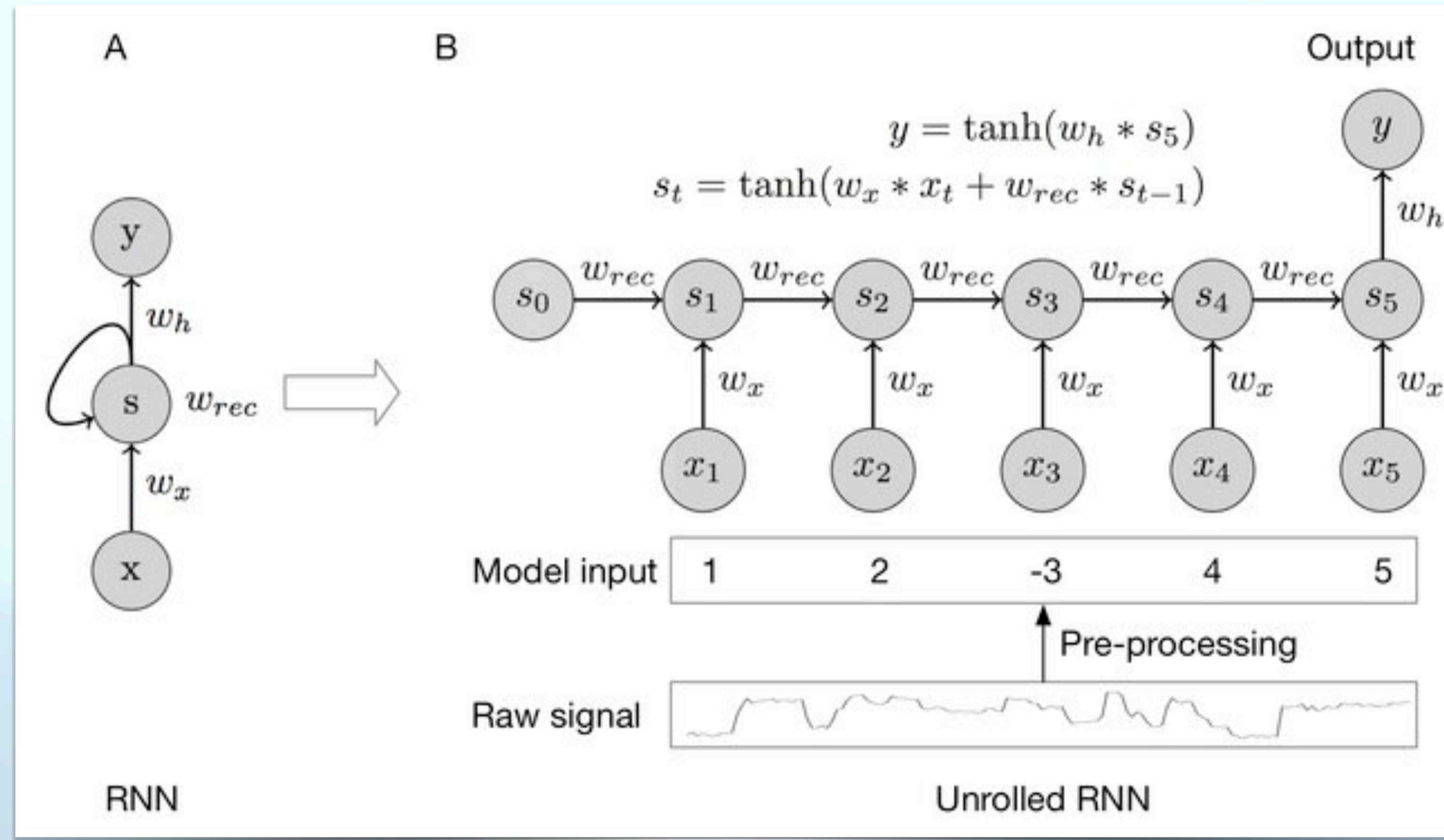
Recurrent Neurons and Layers

$$\mathbf{y}_t = \phi \left(\mathbf{X}_t \mathbf{W}_x + \mathbf{y}_{t-1} \mathbf{W}_y + \mathbf{b} \right)$$



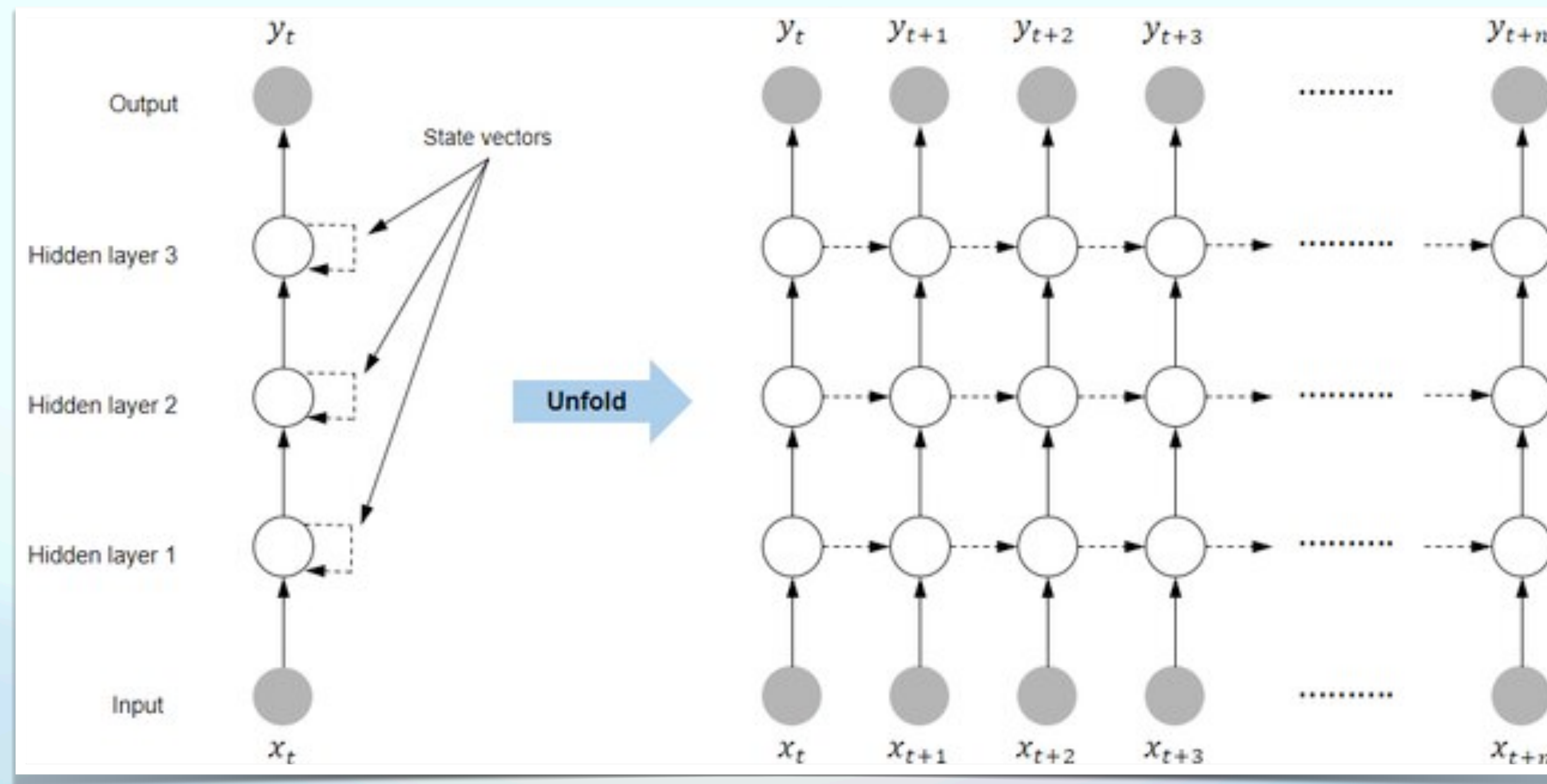
RNN Fundamentals

Handling Temporal Data



RNN Fundamentals

Deep RNN



RNN Fundamentals

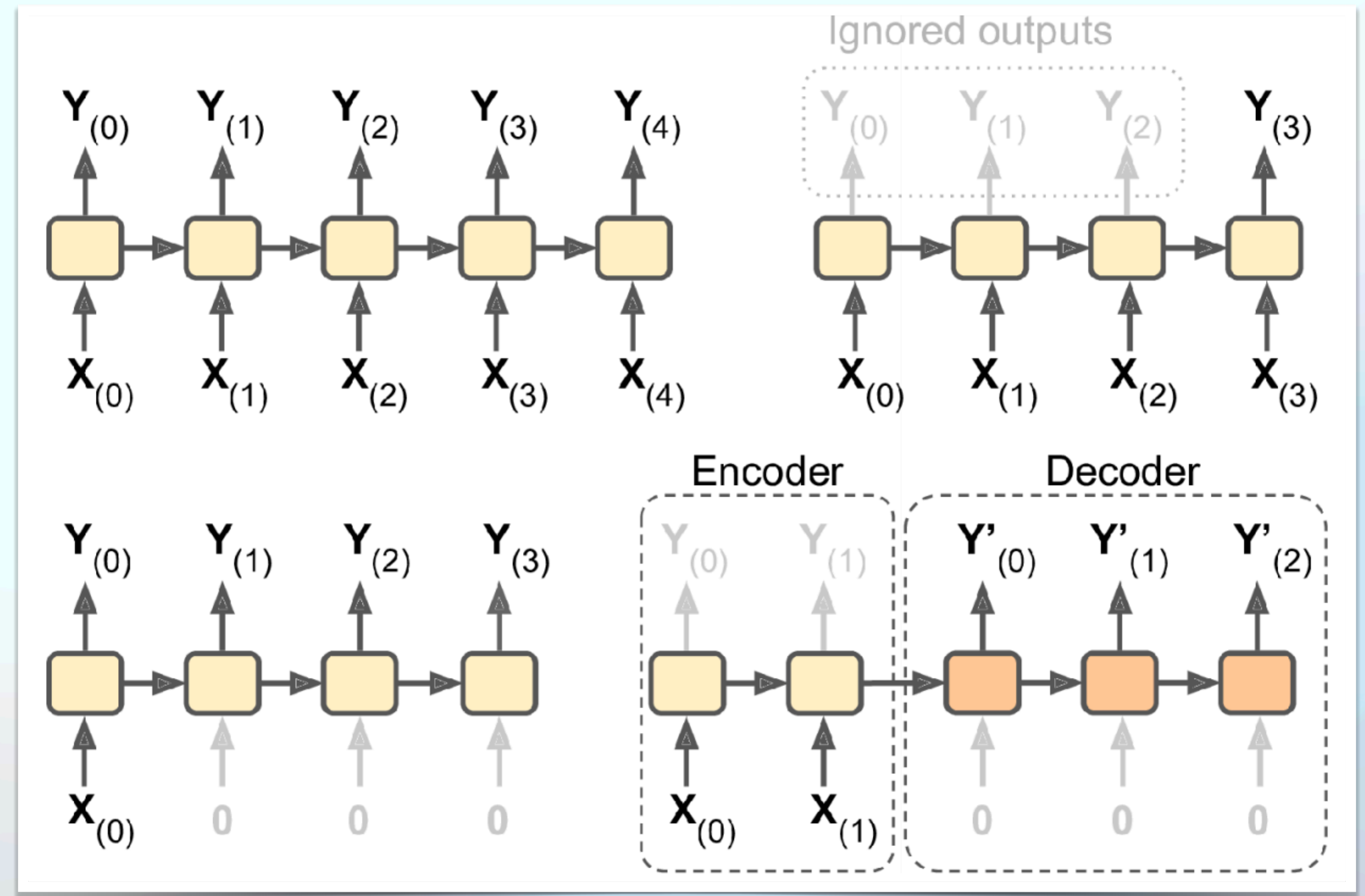
Input and Output Sequences

seq-to-seq (top left)

seq-to-vector (top right)

vector-to-seq (bottom left)

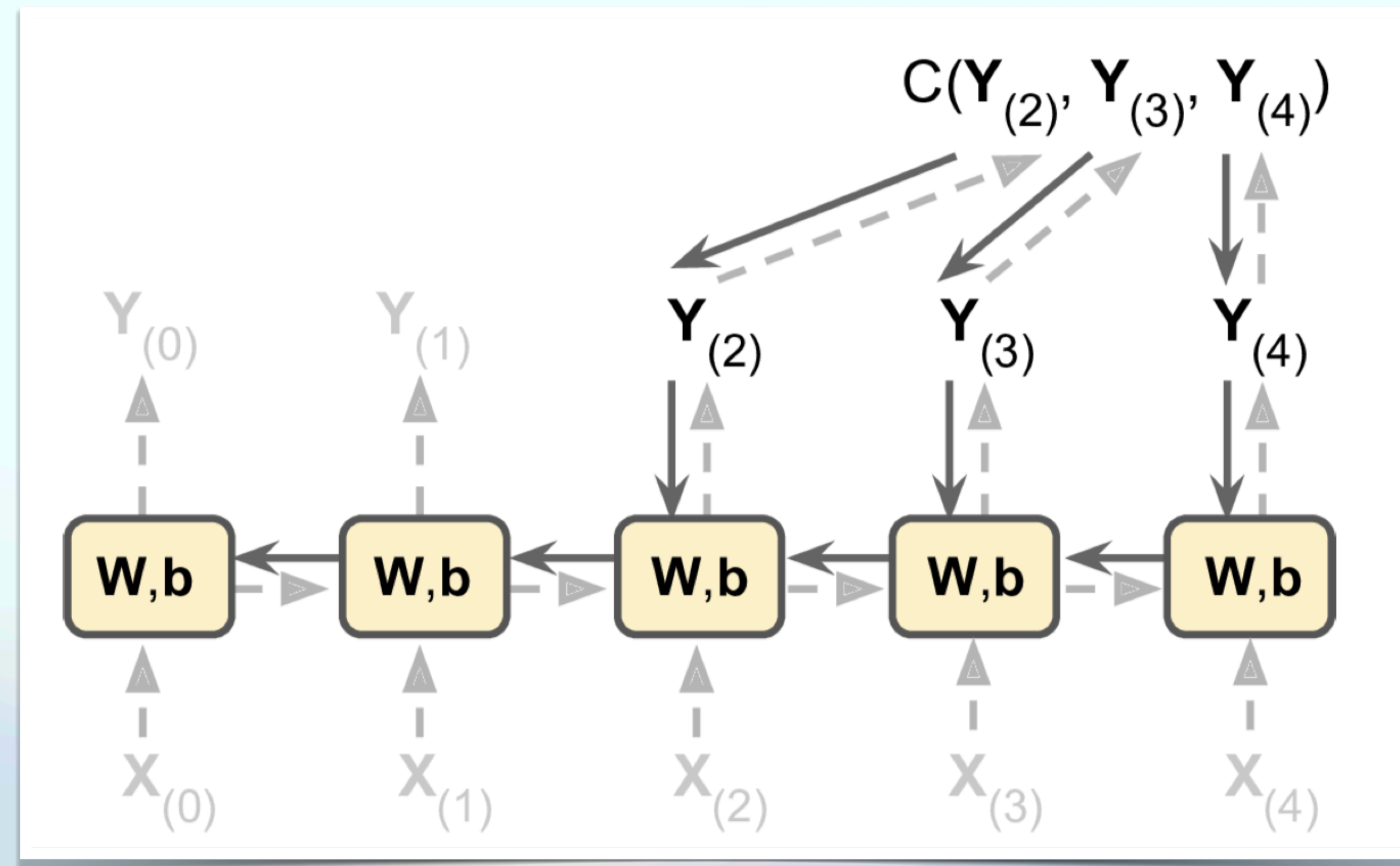
EncoderDecoder (bottom right)
networks



RNN Fundamentals

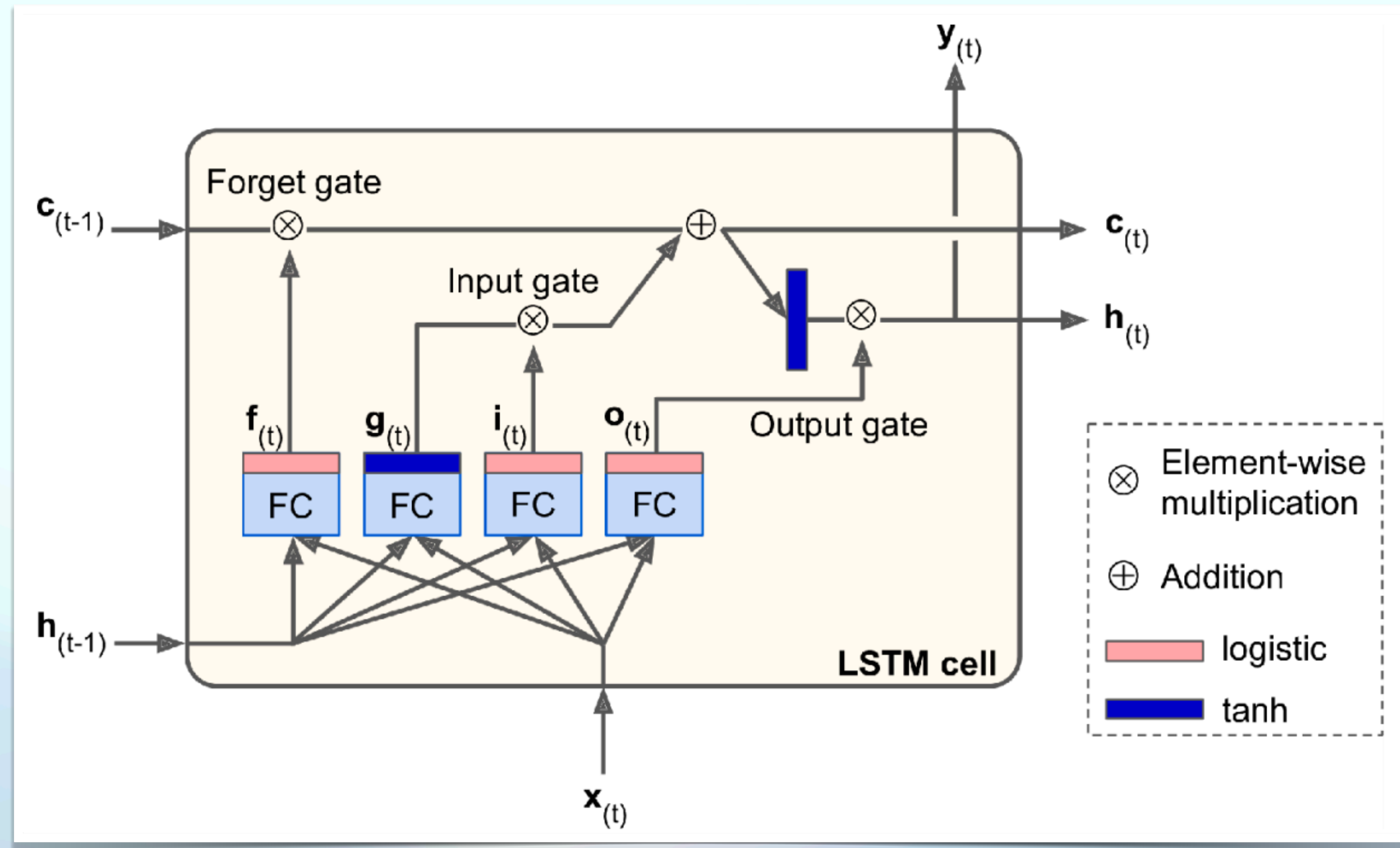
Training RNN

Via backpropagation through time



RNN Variants

LSTM



RNN Variants

GRU

